

SARA SHARMA

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Research Interests

AI Safety · Robust Machine Learning · Model Evaluation · Explainable AI · Machine Learning Reliability · Multimodal AI · AI Governance

Research & Professional Experience

- Indian Institute of Technology Bombay** | *AI/ML Intern* *Sept 2026 – Present*
- Building an end-to-end machine learning research project involving model training, dataset curation, and spatial scene understanding for architectural applications.
 - Working across dataset preparation, model development, and evaluation for spatially structured data.
- NITI Aayog** | *Data Management & Analytics Intern* *Aug 2026 – Present*
- Developed prescriptive machine-learning and forecasting solutions for analysis of complex datasets supporting policy decisions.
 - Translated policy objectives into data-driven analytical solutions through requirements analysis and process evaluation.
- Fluentgrid Ltd.** | *Python & Machine Learning Intern* *June 2025 – Aug 2025*
- Engineered a multi-engine XGBoost system incorporating forecasting, risk analysis and scenario analysis for supply-chain intelligence.
 - Implemented SHAP-based model interpretability to examine model behaviour and support transparent, data-driven decisions.
- Inferrix Greentech** | *AI Research Intern* *Jul 2024 – Sep 2024*
- Developed a Physics-Informed Neural Network (PINN) integrating HVAC, weather, building and real-time IoT data for predictive energy modelling.
 - Incorporated heat-load profiles into the deep-learning architecture, achieving a reported 15% reduction in building energy consumption.

Ongoing Research | Volunteer | Defence Research & Development Organization

- AI/ML Researcher** | DRDO — Scientist-Supervised Research *2026 – Present*
- Developing an AI/ML model for defence applications under the technical guidance and approval of a DRDO scientist.
 - Researching AI workflows integrating heterogeneous data modalities for defence-oriented applications.

Research & Publication

- "ECOAVIA"- AI-Driven Hybrid Forecasting for Sustainable Aviation & Development** | ICISC 2026 [\[Link\]](#)
- Designed an ensemble model (XGBoost, ARIMA, LSTMs) integrating climate and infrastructure data to forecast aviation demand and support sustainable policy advisory.

Projects

- When Should an ML System Not Trust Itself?** | *Evaluating ML Reliability Under Distribution Shift [Python · XGBoost]*
- Developed a framework to evaluate ML reliability under temporal and spatial distribution shift using uncertainty, calibration & OOD detection.
 - Compared standard, uncertainty-aware and abstention-based models to analyze failure modes and risk-performance trade-offs.
 - Extended evaluation to LLMs under adversarial and out-of-distribution inputs.
- AI-Driven Supply Chain Intelligence Platform** *[Python · Scikit-learn · XGBoost · SHAP · Next.js]*
- Engineered a 5-engine XGBoost system (forecasting, risk, scenario analysis) to optimize supply chain visibility.
 - Implemented SHAP analysis to ensure model interpretability, enabling data-driven stakeholder decision-making.

Education

- SRM Institute of Science and Technology** *Chennai, TN*
- Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI & ML) | CGPA: 8.18/10* *2026*

Technical Skills

- **Programming:** Python, SQL, JavaScript
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Neural Networks, Deep Learning
- **Research & Modeling:** Statistical Modeling, Model Evaluation, Anomaly Detection, Explainable AI, SHAP, Time-Series Modeling
- **AI:** Multimodal AI, NLP, RAG, LLM Applications
- **Data:** NumPy, Pandas, PostgreSQL, ETL
- **Engineering:** FastAPI, Docker, Git, CI/CD, MLflow